MINOR PROJECT REPORT

Topic: **Disease Detection in Crops Using Lightweight Deep Learning Models from Images**

Domain: Smart Agriculture

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Acknowledgement

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project.

First and foremost, we extend our heartfelt thanks to our project supervisor, **Dr. Jaspal Kaur Saini** for their invaluable guidance, continuous support, and encouragement throughout the duration of this project. Their expertise and constructive feedback have been instrumental in shaping this work.

We would also like to thank all the faculty members of the Department of Information Technology for their support and valuable suggestions during various stages of this project.

Our sincere thanks go to our family and friends for their constant encouragement and support throughout our academic journey.

Finally, we acknowledge the contributions of all those who directly or indirectly helped us in completing this project successfully.

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Abstract

Crop diseases pose a significant threat to global agriculture by reducing crop yield, affecting food quality, and causing economic losses to farmers. Traditional disease identification methods rely on manual inspection, which is time-consuming, labor-intensive, and dependent on expert knowledge. With advancements in Artificial Intelligence, deep learning-based approaches have emerged as effective solutions for automated plant disease detection using leaf images. However, many high-performance models are computationally intensive and unsuitable for real-time deployment on resource-constrained devices.

This project focuses on developing a lightweight, efficient, and reliable crop disease detection system using deep learning. Lightweight CNN architectures such as SqueezeNet, MobileNetV2, and EfficientNetB0 are employed to balance accuracy and computational efficiency. To enhance feature representation, an attention mechanism (CBAM) is integrated into the model, enabling it to focus on disease-relevant regions of leaf images. Additionally, Bayesian MC-Dropout is incorporated to introduce uncertainty estimation and improve generalization.

The system is trained on the PlantVillage dataset and evaluated using performance metrics such as accuracy, precision, recall, F1-score, specificity, and confusion matrix. Experimental results show that while attention improves feature localization and robustness, Bayesian dropout enhances reliability through uncertainty estimation. The combined approach achieves high classification accuracy (~97–98%) with improved model stability.

The final model is suitable for deployment on mobile and edge devices, making it practical for real-world agricultural applications. This work contributes to the development of intelligent farming systems by enabling early disease detection, reducing dependency on manual inspection, and supporting sustainable agricultural practices.

Contents

1. Introduction

1.1 Overview

This project is presenting a workflow for developing the crop disease detection system using deep learning. It identifies how plant diseases are detected from leaf images using CNN-based architectures. The process begins with dataset preparation where images are resized, normalized, and augmented to improve learning and reduce the effect of the overfitting.

Three lightweight models — SqueezeNet, MobileNetV2, and EfficientNet-B0 — are used as they have balance accuracy and low computational cost, making them suitable for mobile and edge devices. Performance is evaluated using accuracy, precision, recall, F1-score, and confusion matrices for class-wise analysis.

A key enhancement is the integration of Bayesian MC-Dropout after feature extraction. This allows estimation of prediction uncertainty through multiple stochastic forward passes which helps in improving the generalization and stability.

Results are compared to identify the best model in terms of speed, accuracy, and robustness. Overall, the project shows how lightweight deep learning with Bayesian inference and attention layer can provide best agricultural solutions.

1.2 Problem Statement

Detecting crop diseases early is very important for maintaining productivity , but traditional methods rely on manual inspection, which is slow, labor-intensive, and experience-dependent. Early symptoms are often subtle and easily missed from them, especially when diseases show similar patterns.

In many rural areas, limited access to experts leads to delayed diagnosis and in controlling the spread of it. Existing AI solutions often use heavy deep learning models that require high-end GPUs, making them impractical for use in smartphones or in low-power devices.

The central problem addressed is:
To develop a lightweight, accurate, and uncertainty-aware crop disease detection model for resource-constrained environments.

Key challenges include:

- A lot of variations have in situations like lighting, background, and also the leaf texture
- Getting the similar symptoms across diseases
- Overfitting which is lead due to the limited data.
- We have the need for reliable, uncertainty-aware predictions.

1.3 Objectives

1. To develop lightweight deep learning models for plant disease classification:

This project focuses mainly on forming accurate but also computationally efficient models. Heavy CNNs are unsuitable for rural or mobile use as they need resource constrained systems, so lightweight architectures like SqueezeNet, MobileNetV2, and EfficientNet-B0 need to be used.

2. To enhance model reliability using Bayesian MC-Dropout:

Another important objective is to improve the model's ability to generalize and provide trustworthy output to improve generalization and also avoid the overconfident predictions, Bayesian MC-Dropout is applied by keeping dropout active during inference which helps to enable better outputs.

3. To evaluate model performance using standard metrics:

Model performance is assessed using accuracy, precision, recall, and F1-score to understand overall performance and error patterns like false positives and false negatives.

4. To analyze class-wise performance using confusion matrices:

Different plant diseases often have similar visual patterns, and some diseases may be more difficult to identify correctly. To gain deeper insights, the project uses **confusion matrices** to study class-wise performance.
A confusion matrix visually represents the number of correct and incorrect predictions for each disease category. This helps to reveal the patterns such as:

- Diseases which are frequently misclassified.
- Whether certain classes need more training data or not.
- How will the model behave after adding Bayesian dropout or attention layer.

This analysis is critical for improving the model further and ensuring that it performs reliably across all the disease categories and not just the most common ones.

5. To design a model suitable for real-time deployment:

Real-world deployment is a major factor in the agricultural applications and especially in the rural areas where the access to high-end hardware is very limited. Therefore, another thing that is needed is to design a model that functions efficiently on **mobile devices**, **embedded hardware**, or **agriculture robots**.

6. To create a scalable framework for future improvements:

Finally, the project aims to build a framework that can be expanded and integrated into more advanced agricultural monitoring systems in the future. This includes compatibility with:

- **Drone-based crop monitoring** that can be done for large farmlands.
- **IoT sensors** that can automatically capture the environmental data.
- **Field surveillance systems** that continuously scan crops for early symptoms.
- **Mobile or web applications** used by farmers for early protection.

1.4 Scope of the Project

This project defines clear boundaries to ensure that we have focused, practical application and scope for the future improvements and the modification. It aims to build a realistic, deployable, and academically strong crop disease detection system.

1. Model Development

Focuses on building lightweight CNN models like SqueezeNet, MobileNetV2, and EfficientNet-B0. It includes model design, fine-tuning, hyperparameter tuning, and ensuring low memory usage with fast inference for low-power devices.

- Designing the model structure
- Fine-tuning pre-trained models
- Adjusting hyperparameters
- Ensuring low memory usage and fast inference

2. Dataset Usage

Uses the New Plant Diseases Dataset (Augmented).

Covers preprocessing, normalization, resizing, augmentation, and splitting into training and validation sets to ensure balanced and reliable data.

- Loading, preprocessing, and augmenting the dataset
- Normalizing and resizing images for consistency
- Splitting data into training and validation sets
- Ensuring the dataset is balanced and representative

3. Bayesian Integration

A unique part of this project is the integration of **Bayesian MC-Dropout**, which introduces uncertainty estimation into the model. This falls within the scope because the goal is not just to classify but to classify **accuracy**.
The Bayesian integration includes:

- Adding dropout before the final classifier layer
- Keeping dropout active during inference
- Performing multiple stochastic forward passes
- Calculating the mean and variance of predictions

This improves reliability, reduces overfitting, and helps the model handle real-world variations more confidently.

4. Evaluation Metrics

Another important part of the project's scope is the thorough evaluation of model performance using multiple metrics. Instead of depending more on accuracy, the project measures:

- **Precision** - Number of predicted positives are actually correct.
- **Recall** - Number of actual positives were correctly predicted.
- **F1-Score** - Number of balance between precision and recall.
- **Confusion matrices-** Matrice to analyze class-wise strengths and weaknesses.

5. Deployment Readiness

A critical boundary of this project is to make sure that the models are not only accurate but also can be deployed on cloud so more farmers can use this. The scope includes optimizing the system to work on:

- Smartphones
- Edge devices (e.g., Raspberry Pi, Jetson Nano)
- Agricultural robots
- Drone-based crop monitoring systems

This involves reducing model size, lowering computational requirements, and ensuring fast response time so that farmers can use the system instantly in the field.

6. Research Contribution

Finally, the project aims to contribute meaningfully to research in smart agriculture and lightweight AI. This includes:

- Comparing model performance before and after applying Bayesian dropout,Attention Layer and using both together.
- Conducting experiments on multiple model variants
- Making improvements in accuracy, robustness, and reliability
- Providing insights into where Bayesian layers and Attention Layers can improve the performance.

These contributions help other researchers, students, and developers understand how uncertainty-aware lightweight models can benefit agricultural applications.

2. Background

2.1 Introduction

Crop diseases reduce agricultural productivity majorly, causing economic losses and also affecting food security. Traditional disease detection relies on **manual inspection**, which is very **time-consuming, labor-intensive, and dependent on individual expertise**, often leading to delayed and ineffective disease handling.

Advancement in **AI and deep learning** which has enabled automated image-based disease detection with faster and more consistent diagnosis. However, most existing models are **computationally heavy** and require high-performance hardware, making them unstable for **resource-constrained rural areas** . This creates a need for a **lightweight, accurate, and best crop disease detection system** that can operate efficiently on **low-power devices** while maintaining high diagnostic performance.

2.2 Existing Systems and Approaches

Current crop disease detection systems predominantly utilize deep learning architectures such as VGG, DenseNet, ResNet, and conventional Convolutional Neural Networks (CNNs). These models have demonstrated high accuracy when trained on structured datasets such as PlantVillage. However, their performance is often constrained by high computational complexity, large memory requirements, and dependence on Graphics Processing Units (GPUs).

2.3 Modern Deep Learning Approaches

Recent research emphasizes the use of lightweight deep learning architectures that aim to achieve an optimal balance between high accuracy and computational efficiency. Models such as MobileNet, SqueezeNet, EfficientNet, and ShuffleNet are specifically designed for edge and mobile deployment using techniques like depthwise separable convolutions, fire modules, and compound scaling strategies.

2.4 Technological Gaps

Despite advancements in model design and optimization, several technological challenges persist in the domain of crop disease detection:

- The Deep learning models which exist still remain computationally intensive for edge deployment.
- Limited robustness under diverse real-world lighting and background conditions.
- Absence of uncertainty-aware prediction mechanisms in most of the systems.
- Susceptibility to overfitting due to constrained real-world data diversity.
- Difficulty in distinguishing visually similar diseases with high confidence.

These challenges highlight the necessity for developing intelligent systems that combine efficiency, accuracy, and uncertainty awareness for practical agricultural usage.

2.4.1 Neural Networks

Neural Networks are computational frameworks which are inspired by the human brain that consist of interconnected layers of many artificial neurons.

2.4.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are specialized architectures developed in order to process visual data by extracting hierarchical features from the input image. They employ convolutional layers to detect low-level features such as edges and textures, followed by using the pooling layers for dimensionality reduction and fully connected layers for final classification.

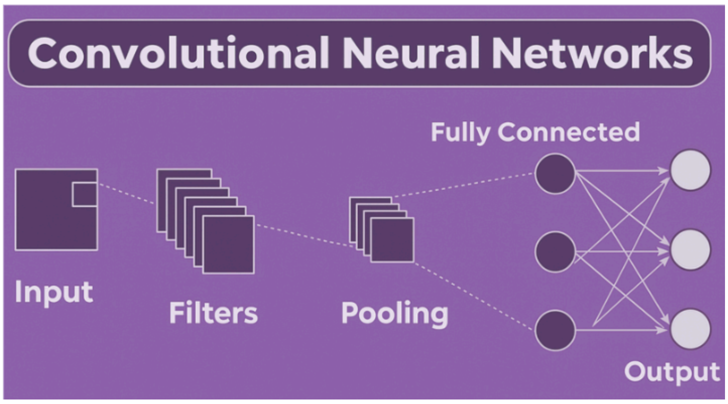


Fig.2.1 CNN Architecture

2.4.3 Bayesian Dropout

Bayesian Monte Carlo Dropout is a probabilistic technique that keeps dropout active during inference and performs multiple forward passes, producing a distribution of outputs which help in estimating predictive uncertainty.

This improves the overall reliability of the model by reducing overconfidence, enhancing generalization, and identifying unclear predictions—especially that are useful in agricultural settings with varying lighting, noise, or occlusion.

2.4.4 Attention Layer

The attention layer is a mechanism designed to enhance the performance of convolutional neural networks by enabling the model to focus on the most important regions of input image. In plant disease detection, not all parts of a leaf contribute equally to classification. Diseased regions such as spots, discoloration, and texture irregularities are more important than background or healthy areas.

The attention mechanism works by assigning higher weights to important features and reducing less useful ones. This allows the network to prioritize disease-specific patterns during feature extraction and classification.

In this project, an attention module (such as CBAM – Convolutional Block Attention Module) is integrated into the CNN architecture. It consists of two main components:
• Channel Attention: It identify which feature maps are important
• Spatial Attention: It identify where important features are located

By combining both channel and spatial attention, the model improves its ability to detect subtle disease patterns, especially in cases where symptoms are small, scattered, or visually similar.The attention layer does not majorly increase computational complexity but improves feature representation quality, making it suitable for lightweight models like SqueezeNet, MobileNetV2, and EfficientNetB0.

Thus, the attention mechanism increases the model's ability to focus more on disease-relevant regions, leading to improved classification performance and interpretability.

Research Gaps

- There is a lack of integration of lightweight CNNs with uncertainty estimation.
- Limited optimization for all the low-powered agricultural devices.
- Poor handling of real-world environmental variation.
- Fewer models have been designed for trustworthy, uncertainty-aware decisions.
- There is a lack of scalable, farmer-friendly systems which are available.

These gaps highlight the need for a solution combining lightweight models with Bayesian inference for accurate, efficient, and reliable crop disease detection.

2.5 Summary

This chapter majorly highlights that although deep learning has greatly improved automated crop disease detection, existing systems remain incomplete for practical deployment in resource-limited agricultural environments. The primary challenges include high computational demands, lack of uncertainty estimation, and insufficient adaptability to these real-world conditions.

3. Motivation

3.1 Introduction

Agriculture is essential for global food production and economic stability, especially in developing countries. But, crop diseases are a major challenge due to the rapid spread and difficulty in early detection, leading to reduced yield, poor quality, and economic losses.

Traditional detection relies on manual inspection, which is time-consuming, subjective, and dependent on expert knowledge. In rural areas, lack of expertise and environmental factors often can make detection inconsistent and worse.

3.2 Need For Automated Crop Disease Detection

Key benefits include:

- Early disease detection before visible symptoms get worse.
- Faster diagnosis which leads to quick decision-making.
- Reduced the dependency on agricultural experts.
- Lower pesticide should be used through needed treatment

Such systems can be deployed using smartphones, drones, or agricultural robots, making them scalable and practical for modern farming.

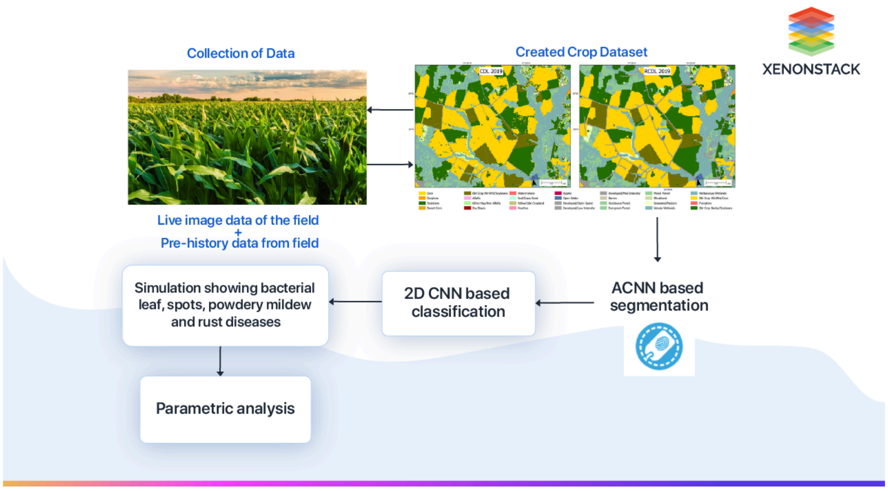


Fig.3.1 Crop Disease Detection Flow

3.3 Importance Of Lightweight Cnn Models

While deep learning models proved highly effective, many popular architectures are too large for real-world agricultural deployment. Lightweight CNN models such as SqueezeNet, MobileNetV2, and EfficientNet are essential because they need less memory, offer faster inference, and consume less power.These models can run finely on mobile phones and embedded devices without cloud connectivity. Their ability to deliver real-time predictions makes them ideal for farmers who need immediate feedback in the field. Ignoring their fact of compact size, they maintain strong accuracy through efficient architectural design.

3.3.1 Technology Stack

The system is designed using a modern and efficient technology stack tailored for deep learning experimentation and deployment:

- **Python**: Primary programming languages for model development.
- **PyTorch**: Deep learning framework used for building and training CNN models.
- **Torchvision**: For data loading, image transformation, and use of pre-trained models.
- **NumPy & Matplotlib**: For data handling, visualization, and also evaluation.
- **scikit-learn**: For calculating precision, recall, F1-score, and generating confusion matrices.
- **GPU/MPS Support**: Utilized to increase training using Mac's Metal Performance Shaders.

3.3.2 Development Environment

The project was implemented in an optimized development environment designed for deep learning experiments:

- **Hardware**: Apple Silicon/MacBook with MPS GPU acceleration
- **Operating System**: macOS
- **Frameworks**: PyTorch + Torchvision
- **IDE**: VS Code / Jupyter Notebook
- **Dataset Structure**: Training and validation images pre-organized in folders
- **Mixed Precision Training**: Used for faster execution and reduced memory use

This environment enabled smooth model training, faster iterations, and experimentation with various CNN architectures and Bayesian techniques.

3.4 Role Of Bayesian Optimization

Bayesian MC-Dropout enhances the reliability of deep learning models by calculating prediction uncertainty. Instead of producing a single fixed output, the model generates multiple predictions, allowing it to assess confidence levels.

1.	**Inside Early Convolution Layers**	**Adding dropout in very early layers affects basic feature extraction**
2.	**In Middle Convolution Blocks**	**Introduces randomness when mid-level abstract features are forming**
3.	**After Feature Extraction**	**Bayesian dropout applied on high-level features**
4.	**After Final Dense Layer**	**Adding randomness before final prediction , not meaningfull**

By introducing randomness during inference, it allows the model to know about how confident it is in disease classification.This uncertainty estimation offers several major benefits:

Preventing Overfitting on Small Datasets

Reduces overfitting on small datasets by introducing randomness during training and inference. This helps the model learn generalized features instead of memorizing patterns.

Making the Model More Confident and Reliable

Enables the model to identify when predictions are confident or uncertain by comparing multiple outputs. This makes the system trustworthy for real-world decision-making.

Producing Smoother Decision Boundaries

Produces flexible decision boundaries instead of rigid ones, which help in improving classification of similar diseases. This is especially useful when classes have overlapping visual features.

Improving Accuracy Through Better Regularization

Extends the effect of dropout into inference, preventing reliance on specific neurons. This leads to more stable and consistent feature learning.

Handling Noisy or Ambiguous Images Effectively

Perform well on real-world images affected by noise, lighting, or occlusion. It reflects uncertainty in unclear cases, making predictions safer and more reliable.

3.5 Role Of Attention Layer

One of the key motivations behind this whole project is to improve the model’s ability to focus on disease-relevant regions in leaf images. In real-world agricultural conditions, images contain noise such as complex backgrounds, lighting variations, shadows, and partial occlusions. Traditional CNN models process the entire image equally, which can lead to reducing the ability to distinguish between important disease patterns from irrelevant information.

The attention layer is introduced to overcome this limitation by enabling the model to selectively concentrate on the most informative parts of the image. Instead of treating all regions equally, the attention mechanism highlights areas showing disease symptoms such as spots, discoloration, and texture irregularities, while suppressing background noise.This becomes especially important in scenarios where:

- Disease symptoms are small or scattered across the leaf.
- Multiple diseases exhibit visually similar patterns.
- Background elements interfere with feature extraction.

Guiding the model's focus, the attention layer improves feature representation and helps in better discrimination between classes. Furthermore, the attention mechanism complements Bayesian optimization used in this project. While Bayesian MC Dropout improves uncertainty estimation and model reliability, the attention layer enhances feature localization. Together, they contribute to a system that is both accurate and interpretable.

1.	Inside Early Convolution Layers	Captures basic features (edges/colors), limited impact.
2.	In Middle Convolution Blocks	Improves texture & pattern understanding.
3.	After Feature Extraction	Best performance, highlights meaningful features.
4.	After Final Layer	Too late, minimal impact.

3.6 Problem Challenges

Developing an accurate disease detection model involves knowing several practical challenges that arise due to variations in real-world agricultural conditions.

1. Noise

Images collected from farms often contain noise due to the inconsistent lighting, shadow, background clutter, or motion blur. This noise could confuse the CNN models and lead to incorrect classification.

2. Similar Diseases

Many plant diseases have visually similar pattern—such as spots, blotches, or leaf discoloration which lead in making them difficult to distinguish. The model must learn subtle differences in texture and shape.

3. Overfitting

Deep learning models could memorize the training images instead of learning general patterns especially when datasets are limited. Overfitting reduces performance on new images.

3.7 Summary

This project aims to develop a lightweight, accurate, and uncertainty-aware crop disease detection system suitable for real-time agricultural environments. Many existing AI-based models are computationally heavy and unstable for low-power devices commonly used by farmers in rural areas. To address this, the project employs lightweight CNN architectures such as SqueezeNet, MobileNetV2, and EfficientNet-B0, which offer high accuracy with minimal computational requirements and can run directly on mobile and edge devices.

These models enable fast image processing and real-time disease detection, allowing timely intervention to prevent disease spread. To improve reliability under real-world conditions—such as noise, uneven lighting, background clutter, and visually similar disease patterns—the project integrates Bayesian MC-Dropout and the attention layer and together both. This technique provides uncertainty estimation, preventing overconfident and potentially incorrect predictions and making the system more trustworthy for farmers.

4. Literature Study

4.1 Introduction

Fast development of deep learning techniques has revolutionized the method of plant diseases recognition, making it transition from visual inspections to computer vision based solutions. Development in the field moved from classical machine learning with artificial features towards fully connected deep neural networks, allowing more efficient recognition of diseases from image datasets.

Our current research work mainly concentrates on improvement of accuracy and efficiency of existing algorithms and methods. In the context of modern trends towards smart farming and agriculture, there is also a considerable attention paid to light weight deep learning models that can be implemented on mobile platforms.

4.2 Review of Existing Research on Plant Disease Detection

Earlier research in plant disease detection mainly used classical image processing techniques combined with machine learning algorithms such as **SVM, KNN, and Random Forests**, relying on manual feature extraction. While these methods performed reasonably in controlled environments, they failed to generalize under varying lighting, background, and leaf orientations.

Recent studies have shifted toward lightweight CNN architectures such as MobileNet, SqueezeNet, EfficientNet, and ShuffleNet, which offer competitive accuracy with fewer parameters. Although techniques like transfer learning and attention mechanisms further improve performance, many existing solutions are still tested on controlled datasets and lack robustness in real-world field conditions, highlighting a gap between research results and practical deployment.

4.3 Comparative Study of CNN Architectures

Deep learning architectures differ significantly in terms of design philosophy, computational requirements, and performance. The following section provides a comparative analysis of widely used CNN models in plant disease detection.

4.3.1 MobileNet

The purpose of MobileNet is to address vision applications for mobile and embedded environments. The model uses depthwise separable convolutions in order to make the computation cost and memory footprint extremely low. It separates the convolution process into two steps (depthwise and pointwise), thus significantly decreasing the number of multiplications performed.

The use of MobileNet for plant diseases classification has proven to yield good results. Moreover, the capability of the model to learn via transfer learning allows for better application of the model in agricultural scenarios.

4.3.2 SqueezeNet

SqueezeNet is a light weight CNN model that is capable of delivering AlexNet-level accuracy using very few parameters. This model uses a fire module composed of squeeze and expand layers, which helps it keep the number of parameters minimal while maintaining the expressiveness of the model.

SqueezeNet provides the benefit of fast inference and compact size making it ideal to be deployed in smartphones; however, due to the limitations in terms of depth, it might fail to capture high levels of complexity in cases of multiple diseases.

4.3.3 ShuffleNet

This architecture has been developed to achieve high speeds of operation while keeping the computational cost low by using group convolutions combined with a shuffle layer. This model has shown good results when applied to high-speed and power-efficient vision applications.

There are many studies reporting good results using ShuffleNet in conjunction with data augmentation techniques or other models such as attention mechanisms when classifying crops' diseases.

4.3.4 YOLO Nano / Tiny YOLO

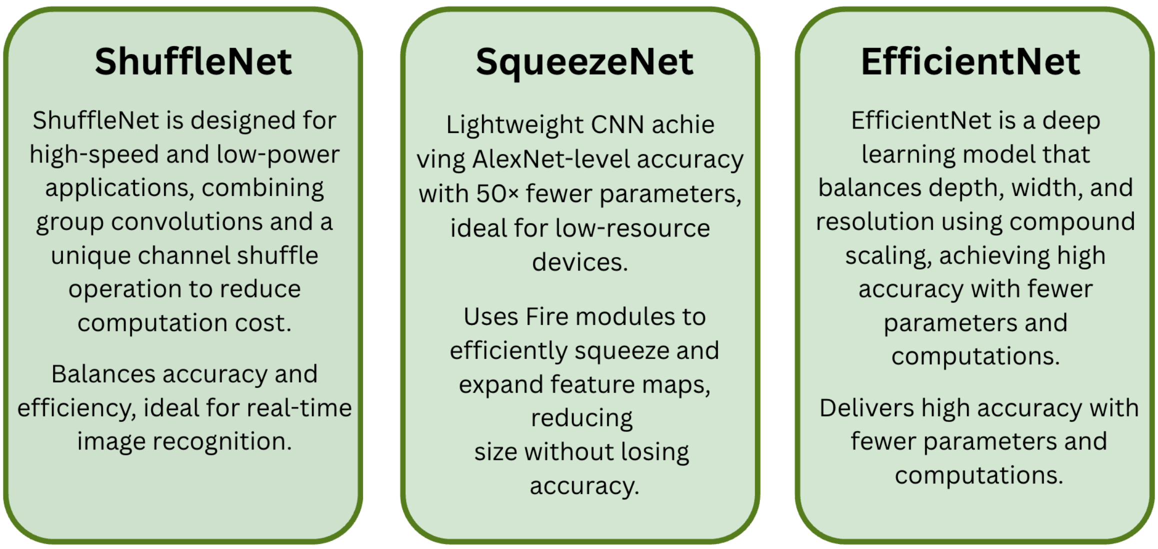
These models are specialized object detectors and they aim at improving the trade-off between speed and accuracy by using simple architectures and reduced number of parameters. They are mostly used in the task of detecting the presence of diseases on crops rather than classification.

Even though they yield impressive real-time performance, they are likely to deliver poor accuracy because of the difficulty to accurately detect tiny regions infected with the disease.

4.3.5 EfficientNet

EfficientNet is a novel architecture based on compound scaling, where network depth, network width, and input resolution are scaled uniformly using a single coefficient. Unlike conventional approaches where increasing one aspect of the network yields a sub-optimal result, this model takes into account all aspects and balances them using a predefined coefficient. EfficientNet-B0 and other versions have been used in many applications including plant disease classification.

It was reported that EfficientNet performs excellently in tasks such as crop disease classification on ImageNet and PlantVillage datasets.



Year	Title	Dataset	Model / Library	Findings	Accuracy	Limitations
2019	YOLO Nano: A Compact You Only Look Once CNN for Object Detection [1]	1. PASCAL VOC 2007/2012 2. Natural images 3. 20 object classes	1. YOLO Nano 2. Human-machine collaborative design 3. PEP architecture 4. FCA attention	1. Balances accuracy and size 2. Efficient embedded deployment	69.1% mAP (VOC 2007) Outperforms Tiny YOLOv2/v3	1. Existing models are large and slow 2. Tiny YOLO sacrifices accuracy
2020	EfficientNet: Rethinking Model Scaling for CNNs [2]	1. ImageNet dataset 2. CIFAR-100, Flowers 3. Transfer learning datasets	1. EfficientNet family 2. EfficientNet-B0 3. MBConv + squeeze-and-excitation	1. Studies compound model scaling 2. Balances depth, width, resolution	Top-1: 84.4% Top-5: 97.1% (ImageNet)	1. Manual tuning required 2. Depends on baseline network
2024	Tiny YOLO Networks for Disease Detection in Paddy Agronomy[3]	1. Rice leaf dataset 2. 15,456 images 3. 3 disease classes	1. Modified Tiny YOLOv4 2. SPP + attention modules 3. UAV-based detection	1. Detects small diseased regions 2. Improved UAV accuracy	mAP: 86.97%	1. Small infection points hard to detect 2. Accuracy limited despite real-time speed
2024	MC-ShuffleNetV2: A Lightweight Model for Maize Disease Recognition[4]	1. 2,725 field images 2. Augmented to 10,845 images	1. MC-ShuffleNetV2 2. CBAM attention 3. Mish activation	1. High accuracy 2. Very low FLOPs	Accuracy: 99.86%	1. Limited disease classes 2. Dataset size constraints
2025	SqueezeNet-Based DL Framework for Accurate Tomato Leaf Disease Diagnosis & Classification[5]	1. PlantVillage (Mendeley Data) 2. 14,531 RGB images 3. 9 diseases + healthy 4. Class imbalance	1. SqueezeNet architecture 2. Transfer learning (ImageNet) 3. SGDM, ADAM, RMSProp 4. MATLAB and Python	1. Efficient deployable model 2. High accuracy with low complexity 3. Training parameter analysis	Overall: 99.91% Healthy recall: 100% F1: 99.42%	1. Multiple diseases on one leaf not handled 2. Limited real-world representation
2025	Optimized classification using EfficientNet-LITE & KE-SVM in diverse environments[6]	1. Uncontrolled Kaggle dataset – 3,076 images (Indonesia) 2. PlantVillage Potato dataset – 2,152 images (controlled)	1. EfficientNet-LITE backbone 2. KE-SVM optimized classifier 3. Channel Attention + 1D LBP 4. Keras, OpenCV	1. Hybrid DL + ML approach 2. Edge-device friendly 3. Improved feature extraction	Uncontrolled: 87.82% Controlled: 99.54%	1. Dataset lacks diversity 2. Weak fusion of lightweight + optimization
2023	Improved MobileNetV2 crop disease identification model for intelligent agriculture [7]	1. PlantVillage dataset 2. 5 crops, 25 diseases 3. 37,572 train / 10,359 test	1. MobileNet-RepMLP 2. RepMLP + ECA attention 3. HardSwish activation 4. PyTorch/Keras	1. Low-resource model 2. 59% fewer params 3. Faster inference	Accuracy: 99.53% Precision/Recall: 99.5% Model: 0.91M	1. Trained on controlled images 2. Real-world gap 3. Leaf-only symptoms
2023	BananaSqueezeNet: Lightweight CNN for diagnosis of banana leaf diseases [8]	1. BananaLSD – 937 images 2. 4 classes 3. Augmented dataset	1. SqueezeNet Fire modules 2. Bayesian optimization 3. Transfer learning	1. Diagnoses 3 diseases 2. Smartphone deployable 3. Good generalization	Overall: 96.25% Size: 4.78 MB Time: 17.84 s	1. Small dataset 2. Action guidance missing
2024	Approach to Lightweighting Backbone Network Models through Quantization and Bayesian Optimization[9]	1. COCO dataset for training and validation 2. 5000 images used for inference 3. Supports autonomous driving scenarios	1. MobileNetV3 Small (baseline) 2. Modified MobileNetV3 with Leaky ReLU 3. Bayesian Optimization 4. Dynamic Quantization (INT8)	1. Quantization reduced size by 50% 2. Accuracy improved by 1% 3. Memory reduced by 97.8% 4. Faster inference	Baseline: Standard MobileNetV3 Improved: +1% after optimization	1. Leaky ReLU increases cost 2. Quantization precision loss 3. Tested only on COCO
2022	Classification of Ear Imagery Database using Bayesian Optimization based on CNN-LSTM Architecture [10]	1. Public Ear Imagery Database 2. 880 otoscopic images 3. 4 classes 4. 420×380 resolution	1. EfficientNetB0 (CNN) 2. BiLSTM classifier 3. CNN–LSTM hybrid 4. MATLAB 2020	1. CNN-LSTM outperformed CNN 2. Bayesian optimization improved tuning 3. Training time reduced (25 vs 637 min)	100% testing accuracy CNN-only: 86.3% Sensitivity: 100%	1. Small dataset 2. Single database 3. Not tested clinically
2024	Enhanced MobileNet for Skin Cancer Image Classification with Fused Spatial Channel Attention	1. ISIC-2019 dataset 2. 8 classes of skin lesions 3. Images resized to 224×224 4. Class imbalance present	1. MobileNet-based CNN 2. MobileNet-MFS (Multi-scale fused spatial-channel attention) 3. Compared with SE, CBAM, ECA, CA	1. Improves global + local feature learning 2. Better precision, recall, F1-score 3. Handles long-range dependencies	Peak: 87.3% Training: ¿90% Loss: 0.5–0.6	1. Class imbalance affects results 2. Lower melanoma performance 3. Slightly higher computation 4. Requires large dataset
2024	Machine Fault Diagnosis using Attention Mechanisms with Lightweight SqueezeNet	1. MIMII dataset 2. ToyADMOS dataset 3. Acoustic sensor data 4. 1D → 2D spectrogram conversion	1. Lightweight SqueezeNet 2. 2. Self-attention, CBAM, channel & spatial attention 3. Texture-based preprocessing 4. t-SNE visualization	1. Self-attention performs best 2. Very low parameters (0.08M) 3. Effective audio feature extraction	97% (self-attention) Others: 95–96%	1. Limited real-world variability 2. Depends on preprocessing 3. Sensitive to noise
2022	Efficient Attention-based CNN Network (EANet) for Multi-class Maize Disease Classification	1. Multiple maize datasets 2. Real-world noisy images 3. Class imbalance handled via focal loss	1. EfficientNetV2 backbone 2. Spatial + channel attention 3. End-to-end CNN (EANet) 4. Transfer learning	1. Focuses on disease regions 2. Handles noisy backgrounds 3. Better generalization 4. Reduces class imbalance	99.89% overall accuracy	1. High dataset dependency 2. Complex real-world scenarios 3. Computationally heavier
2017	Attention Is All You Need (Transformer Model)	1. WMT 2014 EN-DE dataset 2. WMT 2014 EN-FR dataset 3. Text sequence data	1. Transformer (encoder-decoder) 2. Fully attention-based 3. Multi-head self-attention 4. Scaled dot-product attention	1. Captures long-range dependencies 2. Faster training via parallelization 3. State-of-the-art performance 4. Basis for BERT, GPT	BLEU: 28.4 (EN-DE) BLEU: 41.0 (EN-FR) ¿2 BLEU improvement	1. Requires large datasets 2. High memory usage ($O(n^2)$) 3. Inefficient for long sequences

4.4 Bayesian Neural Networks in Image Classification

Bayesian Neural Networks apply probabilities in deep learning because their parameters can now be assumed as probability distributions instead of single points. Monte Carlo (MC) Dropout is one of the widely used Bayesian algorithms for computer vision that uses the dropout process in prediction time.

Using this algorithm helps us measure predictive uncertainty, meaning how sure the model is about its prediction. This is especially critical in agriculture because wrong predictions could result in applying unnecessary pesticides.

4.5 Attention Layer in Image Classification

Attention mechanisms address one of the key limitations identified in the literature—lack of focus on disease-specific regions in images.

While traditional CNNs process the entire image uniformly, attention-based models dynamically emphasize important features and suppress irrelevant ones. This is particularly useful in agricultural datasets where:
• Backgrounds may contain noise or clutter
• Disease regions may be small or partially visible
• Multiple visual patterns may overlap

By integrating attention layers, the model:
• Improves feature localization
• Enhances class separability
• Reduces misclassification in visually similar diseases

However, based on experimental trends, attention does not drastically improve accuracy when baseline models are already highly optimized. Instead, it contributes to:
• Better feature representation
• Improved robustness
• Enhanced interpretability

Thus, attention mechanisms complement CNN architectures by refining how features are processed rather than fundamentally changing model capacity.

4.6 Identified Limitations in Literature

Based on the reviewed studies, several recurring limitations have been identified:

- Heavy reliance on controlled datasets such as PlantVillage, limit generalizability
- High computational requirements of many deep learning models
- Lack of uncertainty estimation in most classification systems
- Insufficient evaluation under changing environmental conditions
- Difficulty in detecting subtle or visually similar disease symptoms

These limitations highlight the necessity for a more practical approach that addresses both performance and deployability.

Comparison:

This study provides a comparative analysis of deep learning algorithms that can be used for detecting diseases in crops using transfer learning, paying particular attention to their suitability for rural areas where resources are scarce. We have developed a comprehensive dataset by applying several preprocessing techniques to data collected from PlantVillage and New Plant Diseases Datasets.

We show how transfer learning helps to achieve better accuracy and robustness of models. After comparing different models and considering accuracy as well as performance measures, we found EfficientNet to be a winning algorithm because of its perfect balance of accuracy and computation complexity. Mobile-compatible, off-line enabled models are the main focus of our research.

4.7 Summary

Progress has already been made in terms of plant disease detection with deep learning algorithms. Despite the great success of CNN-based architectures, current approaches still suffer from high computation cost, lack of awareness about model certainty, and inability to work in the real world.

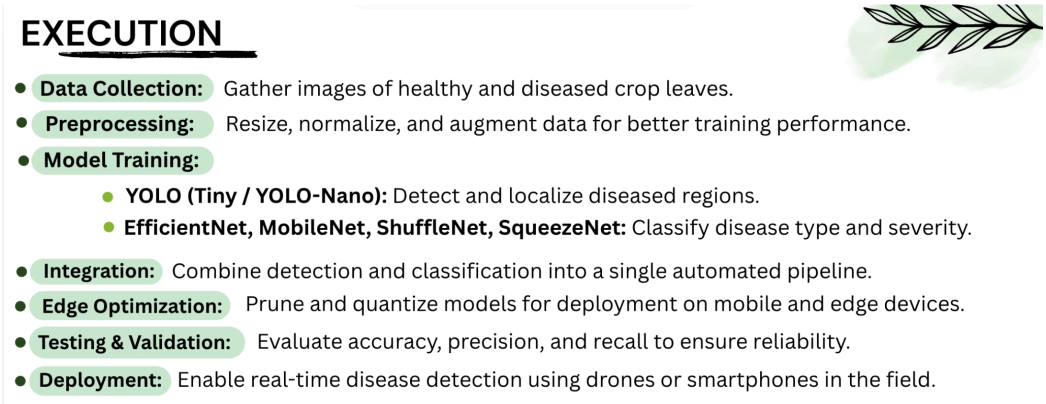
However, there are promising solutions such as Lightweight Convolutional Neural Networks, which can be deployed on the edge devices. The incorporation of a Bayesian approach is also very helpful.

5. <u>Project Objectives</u>

5.1 Introduction

The following part explains the experimental design used in creating, training, and assessing the crop disease recognition system. The main goal is to create lightweight DL systems appropriate for real-time applications in agriculture. This section addresses the problem of the experimental design, including the creation of datasets, modeling and the computational hardware used.

It includes computing hardware details, preprocessing of datasets, implementation of models such as SqueezeNet, MobileNetV2, EfficientNet-B0, and use of Bayesian MC-Dropout for better generalization and uncertainty estimation. Evaluation includes the calculation of relevant performance metrics and comparing before and after adding Bayesian dropout.



5.2 Experimental Setup

The key components of this setup include:

- **Model Preparation:** Lightweight CNN architectures are initialized using pre-trained weights (ImageNet) and then fine-tuned on the plant disease dataset.
- **Bayesian MC-Dropout Integration:** A dropout layer is added before the classifier to enable uncertainty-aware predictions and reduce overfitting.
- **Training Infrastructure:** PyTorch is used as the primary deep learning framework, utilizing GPU/MPS acceleration for fast computation.
- **Evaluation:** After each epoch, validation is performed, and metrics such as accuracy, precision, recall, and F1-score are computed.
- **Confusion Matrices:** Detailed class-wise evaluation helps identify which diseases are classified accurately and where misclassifications occur.

5.2.1 Hardware Configuration

To develop and train the crop disease detection models, a carefully configured hardware environment was used.

- **Device:** Apple MacBook Air / MacBook running Apple Silicon
- **Processor:** 8-core CPU optimized for parallel tasks

- **GPU Acceleration:** Metal Performance Shaders (MPS) providing GPU-level speed for PyTorch training
- **Memory:** 8–16 GB Unified Memory, allowing fast data sharing between CPU and GPU
- **Storage:** SSD for quick dataset loading, checkpoint saving, and model retrieval
- **Energy Efficiency:** Lightweight models reduced power usage, enabling long-duration experiments without overheating or throttling

5.2.2 Software Configuration

The software ecosystem used in this project ensures a flexible yet powerful development environment.

- **Programming Language:** Python 3 for its simplicity and extensive support in deep learning
- **Deep Learning Framework:** PyTorch for building neural networks, fine-tuning pre-trained models, and handling GPU computation
- **Torchvision:** For image preprocessing, transformations, and dataset loading
- **Supporting Libraries:**
    - **NumPy:** Array operations and data manipulation
    - **Matplotlib:** Visualizations, plotting accuracy/loss graphs
    - **scikit-learn:** Precision, recall, F1, confusion matrix
- **Operating System:** macOS with MPS backend
- **IDE:** Visual Studio Code and Jupyter Notebook for coding, debugging.
- **Training Enhancements:** Mixed-precision training (float16) to improve speed and reduce memory usage

5.3 DATASET DESCRIPTION

The **New Plant Diseases Dataset (Augmented)** is used for training and validation. It is one of the largest and most diverse datasets for plant disease recognition.

Key dataset features:

- **Image Format:** High-resolution RGB leaf images
- **Total Classes:** Approximately 38, including both diseased and healthy categories
- **Structure:**
    - **Train Folder:** Contains thousands of class-wise augmented images
    - **Validation Folder:** Contains separate images never seen during training
- **Variations Covered:**
    - Lighting changes
    - Color intensity variations
    - Background differences
    - Rotation, flip, and zoom augmentations

5.4 MODEL ARCHITECTURE OVERVIEW

The project uses three lightweight CNN models:

1. SqueezeNet

- Utilizes “Fire Modules” (squeeze + expand layers)

- Extremely small model size (~1.2 million parameters)
- High speed and low memory usage

2. MobileNetV2

- Uses depthwise separable convolutions
- Very efficient in computation with high accuracy
- Highly suitable for mobile and IoT devices

3. EfficientNet-B0

- Uses compound scaling to balance network width, depth, and resolution
- Provides strong accuracy despite being lightweight
- High performance-to-parameter ratio

These architectures allow the system to run efficiently on low-power hardware such as smartphones and agricultural devices

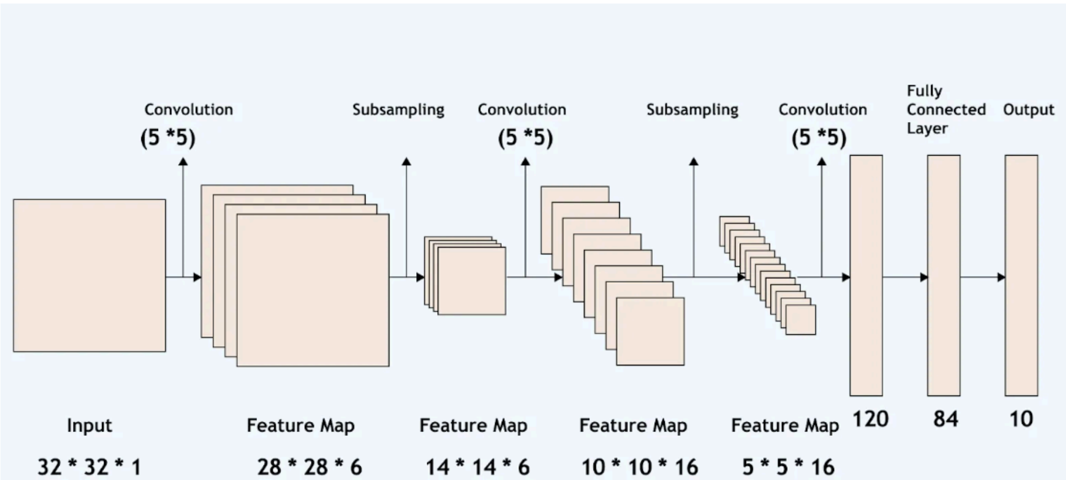


Fig.5.1 Architecture of the CNN Models

5.5 BAYESIAN LAYER PLACEMENT STRATEGY

The Bayesian MC Dropout algorithm is applied in this project as a means of providing an element of uncertainty in the prediction phase of the algorithm and as a way of improving the generalization capacity of the lightweight CNN model. However, the location of the Bayesian dropout layer in the design is very important since wrong positioning may affect either the extraction of features or the model's efficiency.

In this project, the Bayesian dropout layer is placed:

- ➔ **After the final feature extraction block**
- ➔ **Immediately before the final classifier layer**

This placement is carefully selected based on the behavior of CNN architectures and the intended role of Bayesian inference. The reasoning behind this strategy is explained below in detail:

1. Early Convolution Layers Must Remain Stable

The first few layers of any CNN learn **low-level features** such as edges, corners, textures, and color gradients. These features form the foundation for all deeper-level understanding of the image.

If dropout were applied too early in the network:

- Important low-level information would be lost randomly
- Feature maps would become inconsistent
- The model would struggle to learn basic patterns
- Accuracy would drop significantly

2. Bayesian Dropout Works Best on High-Level Representations

As the network progresses deeper, it begins to learn **high-level semantic features**, such as:

- Specific disease patterns
- Texture irregularities
- Spot shapes and distributions
- Color distortion caused by infections

These higher-level features are **more abstract** and **more sensitive to uncertainty**. Applying Bayesian dropout at this stage provides meaningful randomness that helps the model capture uncertainty in ambiguous or noisy samples.

3. Enhances Classifier Robustness Without Affecting Feature Learning

The classifier layer is responsible for mapping extracted features into specific disease labels.
By adding Bayesian dropout just before this classifier, the network:

- Becomes more resistant to overfitting
- Learns more generalized feature-to-class relationships
- Avoids memorizing training data patterns
- Produces smoother and more flexible decision boundaries

4. Enables Uncertainty Estimation via Multiple Stochastic Forward Passes

Placing the dropout near the classifier ensures that this uncertainty directly reflects:

- Ambiguity in disease patterns
- Noisy or low-quality images
- Visually similar diseases
- Edge cases that the model is unsure about

This leads to more trustworthy predictions.

5. Reduces Overfitting by Regularizing Only the Final Decision Layers

Overfitting is common in agricultural datasets due to variations in lighting, texture, and background.
Bayesian dropout:

- Adds randomness
- Prevents memorization
- Encourages generalization

By applying it only at the final layer, the model:

- Learns stable feature maps
- Regularizes only the part prone to overfitting
- Achieves better validation accuracy
- Shows stronger performance on unseen images

5.6 ATTENTION LAYER PLACEMENT STRATERGY

In this particular project, the attention layer has been employed in order to increase the efficiency of the network in terms of identifying the relevant parts of the plant leaf image, especially those parts that are affected by the disease. Just like Bayesian dropout, however, the position at which the attention layer has been included in the architecture of the CNN also plays a major role.

Therefore, the attention module must be strategically positioned to maximize its contribution to feature refinement while preserving the integrity of the feature extraction process.

In this project, the attention layer is placed:
• After the final feature extraction block (CNN backbone output)
• Before the global pooling and classification layers

This placement is carefully selected based on the hierarchical nature of CNN feature learning and the functional role of attention mechanisms. The reasoning behind this strategy is explained below:

Early Convolution Layers Should Not Be Disturbed

The initial layers of a CNN capture low-level features such as edges, textures, and color gradients. These features are generic and form the foundation for deeper representations.

If attention is applied too early:
• It may suppress essential low-level features
• Feature extraction becomes unstable
• The network may fail to learn basic visual patterns
• Overall model performance may degrade

Thus, early layers are kept free from attention to maintain stable feature learning.

Attention Works Best on High-Level Semantic Features

As the network progresses deeper, it learns more meaningful representations such as:
• Disease-specific patterns
• Texture irregularities
• Color distortions caused by infections
• Shape and spread of infected regions

These high-level features are ideal for attention mechanisms. Applying attention at this stage allows the model to:
• Highlight disease-relevant regions
• Suppress background and irrelevant information
• Improve discrimination between similar classes

Preserves Spatial Information for Effective Localization

Before global pooling, the feature maps still retain spatial structure (height × width × channels).

Attention requires this spatial information to:
• Identify where important regions are located

• Assign importance to specific locations in the image

If attention is applied after pooling:
• Spatial information is lost
• The model cannot localize disease regions
• The attention mechanism becomes ineffective

Enhances Feature Refinement Before Classification

By placing attention just before the classifier:
• The feature maps are refined and re-weighted
• Important features are amplified
• Less useful features are suppressed

This leads to:
• Improved feature quality
• Better class separability
• More accurate predictions

Avoids Late Placement Where Impact Is Minimal
If attention is applied after the final dense layer:
• The classification decision is already made
• There is no meaningful feature refinement possible
• The impact of attention becomes negligible

Thus, placing attention too late in the network provides little to no benefit.

Improves Robustness Without Increasing Complexity Significantly
The chosen placement allows attention to:
• Enhance feature representation
• Improve robustness against noise and background variations
• Maintain computational efficiency

This makes it suitable for lightweight CNN architectures used in this project.

Conclusion of Placement Strategy

The attention module works best when used post-feature extraction but before classification. It allows it to work on a higher level representation without losing spatial information, making its function more focused, robust, and comprehensible.

5.7 TRAINING, TESTING, AND VALIDATION

Training Phase

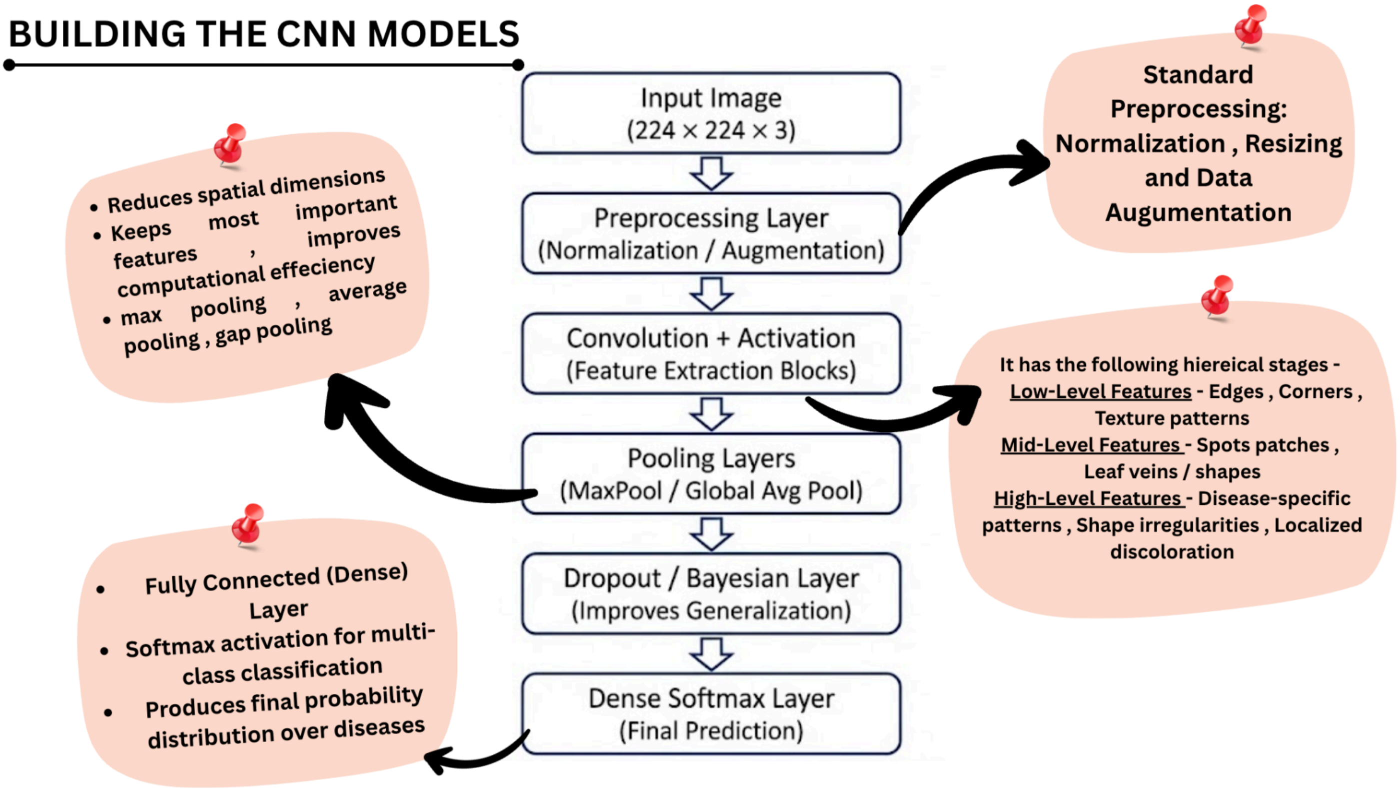
- Models trained for **5 to 20 epochs** depending on architecture
- Batch size set to **32**
- Optimizer: **Adam** with learning rate **0.0007**
- Loss function: **CrossEntropyLoss**
- Mixed-precision training used to make training faster
- Dropout applied to reduce overfitting

Testing Phase

- Performed on validation data
- Gradients disabled (torch.no_grad())
- Bayesian dropout kept active for uncertainty estimation
- Predictions collected for each batch

Validation Phase

- Accuracy computed after every epoch
- Predictions stored for calculating precision, recall, and F1-score
- Confusion matrices generated to evaluate class-wise performance
- Both **before** and **after** Bayesian dropout results compare.



5.8 EVALUATION METRICS

To achieve this, the project uses a combination of four widely accepted metrics: **Accuracy, Precision, Recall, and F1-score**. Each metric captures a different aspect of the model's behavior and helps give a complete picture of overall system performance.

5.8.1 Accuracy

Accuracy is the most straightforward metric used to evaluate classification models. It measures the proportion of correct predictions out of the total number of predictions made by the model.

Why Accuracy Matters

- Provides a **broad overview** of how well the model performs across all classes.
- Helps compare the general performance of different models (SqueezeNet, MobileNetV2, EfficientNet).
- Useful for understanding model improvements before and after adding Bayesian MC-Dropout.

Interpretation in Agriculture Context

A high accuracy score indicates that the model is correctly classifying the majority of leaf images, making it effective for real-time disease screening in fields.

5.8.2 Precision

Precision measures how many of the samples predicted to belong to a certain class actually belong to that class. It focuses on the correctness of **positive predictions**.

Why Precision is Important

- Minimizes **false positives**, where a healthy plant is mistakenly classified as diseased.
- Avoids unnecessary pesticide use, saving cost and preserving plant health.
- Crucial when certain diseases look visually similar; precision ensures the model is careful and selective.

Interpretation in Agriculture Context

High precision means the model is reliable when it predicts a disease — it rarely raises false alarms.

5.8.3 Recall

Recall measures how many actual positive samples the model correctly identifies. It focuses on the ability to **detect true disease cases**.

Why Recall is Important

- Reduces **false negatives**, where diseased plants are mistakenly classified as healthy.
- Critical for preventing disease spread, especially in fast-spreading infections like blight or mildew.
- Ensures that the model does not overlook early-stage symptoms.

Interpretation in Agriculture Context

High recall ensures that the model successfully identifies the majority of infected plants, even if symptoms are subtle.

5.8.4 F1-Score

The F1-score is the harmonic mean of precision and recall. It combines both metrics into a single value that reflects a balanced evaluation of the model's performance.

Why F1-Score is Important

- Useful when classes in the dataset are imbalanced.
- Provides a more realistic measure of performance than accuracy alone.
- Helps understand whether the model is both **accurate** and **consistent**.

Interpretation in Agriculture Context

A high F1-score means the model:

- Detects diseased leaves accurately (high recall)
- Makes correct disease predictions with minimal confusion (high precision)

SNO.	PARAMETER	VALUE
1.	Epochs	2–20 depending on experiment
2.	Batch Size	32
3.	Metrics	Accuracy, Precision, Recall, F1-score
4.	Callbacks	ModelCheckpoint + EarlyStopping

6. UML DIAGRAMS

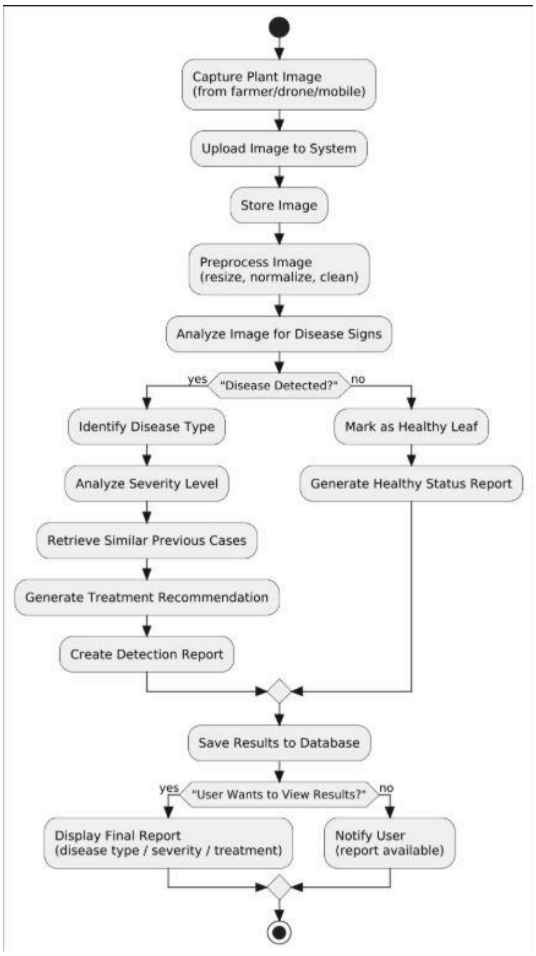
6.1 INTRODUCTION

UML diagrammatic representations are crucial for understanding, analyzing, and depicting the structure of a software system. UML diagrams are important for this project since they enable the visualization of how various components of the crop disease detecting system communicate and interact. As such, UML enables us to depict data flow in the software by illustrating how various modules within the system communicate with each other. Since the project entails machine learning and system deployment, among others, UML helps to easily depict the software structure.

By using UML diagrams, the project ensures:

- **Better clarity** in system design
- **Easy communication** of architecture
- **Well-organized documentation** for academic evaluation
- **Understanding of interactions** between software modules and machine-learning components
- **Improved maintainability**, making the system easier to extend in future work (e.g., adding drones, IoT sensors, or new models.

ACTIVITY DIAGRAM



7. Experimental Results

7.1 Introduction

The implementation of the experiment for testing the effectiveness of the crop disease detection system with the EfficientNetB0, MobileNetV2, and SqueezeNet models is presented in this chapter. All the three models were trained on the PlantVillage dataset. Performance analysis of all three models was conducted both before and after employing the Bayesian optimization technique with MC Dropout. The experiments involved comparing accuracy, creating confusion matrices, analyzing precision, recall, and F1 scores, and determining the most effective location to apply the Bayesian dropout layer.

7.2 Model Training Results

All three models—EfficientNetB0, MobileNetV2, and SqueezeNet—were trained using a 90:10 train-validation split. Key observations include:

- EfficientNetB0 achieved the highest validation accuracy among all models due to its compound scaling design
- SqueezeNet, despite its very small size, performed surprisingly well when regularized with Bayesian dropout.
- MobileNetV2 benefited from improved calibration, although its raw accuracy slightly varied under aggressive dropout rates.

SQUEEZENET

Layer (type)	Output Shape	Param #
input_layer	(1, 3, 224, 224)	0
squeezenet_backbone	(1, 512, 13, 13)	722,496
global_avg_pool	(1, 512)	0
dropout	(1, 512)	0
output_dense	(1, 15)	7,695

Fig.7.1 Layer-wise architecture of the proposed SqueezeNet model

EFFECIENTNET

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
efficientnetb0 (Functional)	(None, 7, 7, 1280)	4,049,571
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 15)	19,215

Fig.7.2 Layer-wise architecture of the EfficientNet-B0 model

MOBILENET

Layer (type)	Output Shape	Param #
input_1	(None, 224, 224, 3)	
mobilenetv2_1 (Functional)	(None, 7, 7, 1280)	0
global_average_ (GlobalAveragePooling2d)	(None, 1280	0
dropout	(None, 1280	0
dense	(None, 15	19,335

Fig.7.3 Layer-wise architecture of the MobileNetV2 model

7.3 Confusion Matrix (Before & After Bayesian Optimization)

Confusion matrices were generated for each model to analyze class-wise performance on the 15 crop categories. Before Bayesian optimization, certain visually similar diseases (e.g., Early Blight vs. Late Blight) showed classification confusion.

After introducing the Bayesian dropout layer:

● EfficientNetB0 and SqueezeNet displayed much clearer diagonals in the confusion matrix, indicating improved class precision.
● Misclassifications reduced for diseases with subtle visual differences.
● MobileNetV2 showed more calibrated outputs even when accuracies varied.

SQUEEZE NET

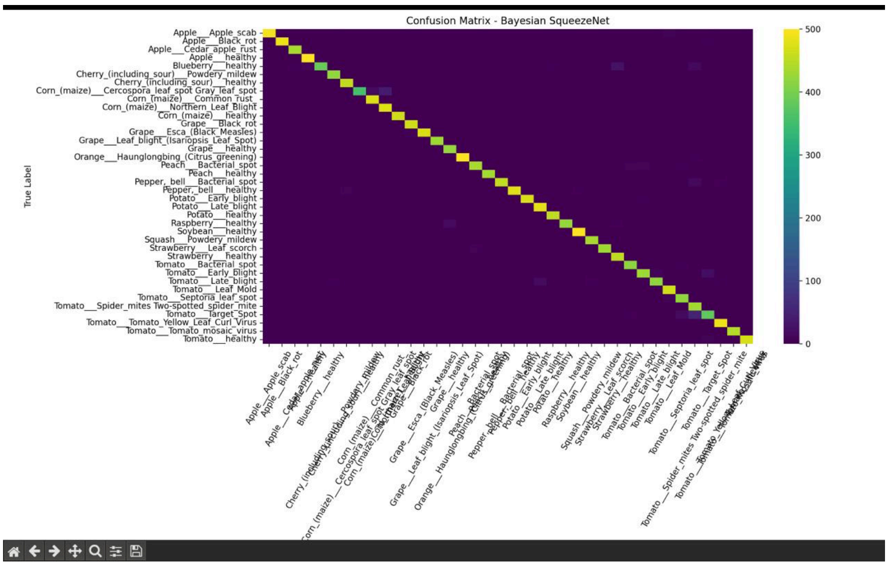


Fig.7.4 Confusion matrix of the Bayesian SqueezeNet model evaluated on the Plant Diseases Dataset

EFFECIENTNET

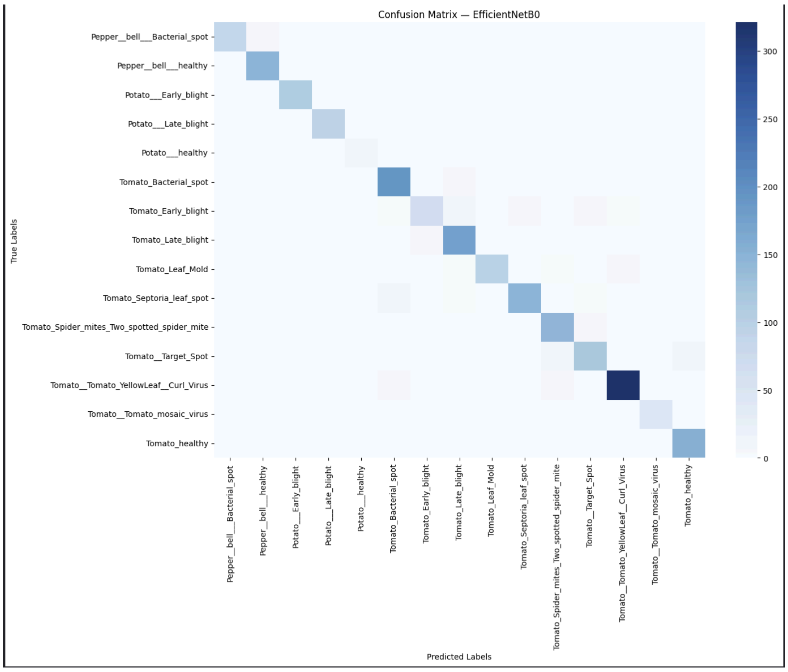


Fig.7.5 Confusion matrix of the EfficientNet-B0 model evaluated on the Plant Village Dataset

MOBILE NET

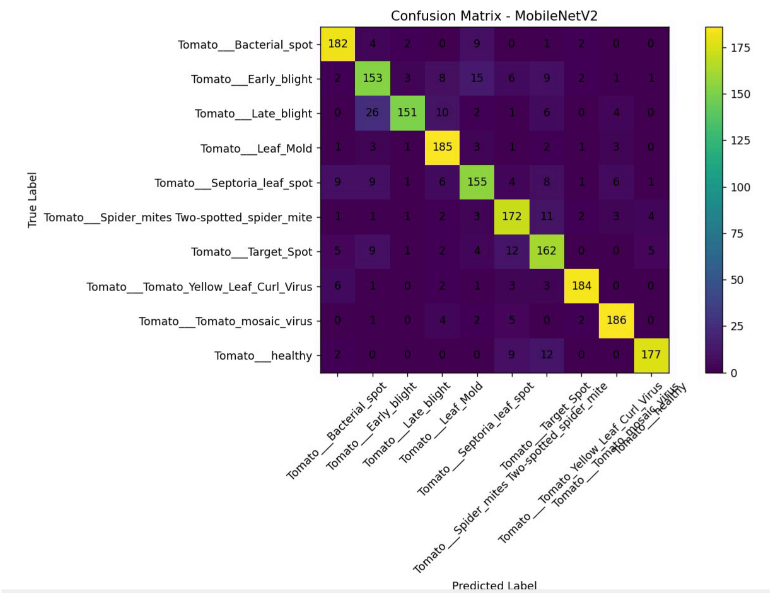


Fig.7.6 Confusion matrix of the MobileNetV2 model evaluated on the Tomato Leaves Dataset

7.4 Precision, Recall, F1-score - :Comparison

General trends observed:

- EfficientNetB0 saw significant improvement in precision and recall, indicating stronger correct classifications across all classes.
- SqueezeNet's F1-score increased due to better handling of minority classes.
- MobileNetV2 exhibited stable recall and improved consistency, despite a minor reduction in accuracy.

SQUEEZENET

Metric	Before	After(Bayesian)
ACCURACY	87%	97%
PRECISION , RECALL	88.74% , 87.10%	97.14% , 97.03%
F1 - Score	87.12%	97.01%

EFFECIENTNET

Metric	Before	After(Bayesian)
ACCURACY	79.45%	92.92%
PRECISION , RECALL	77.25% , 79.39%	85.86% , 85.55%
F1 - Score	77.03%	84.8%

MOBILENET

Metric	Before	After(Bayesian)
ACCURACY	85.40 %	78.10 %
PRECISION , RECALL	88%, 85%	80%, 78%
F1 - Score	86%	79%

7.5 Training V/S Validation Loss and Accuracy curves

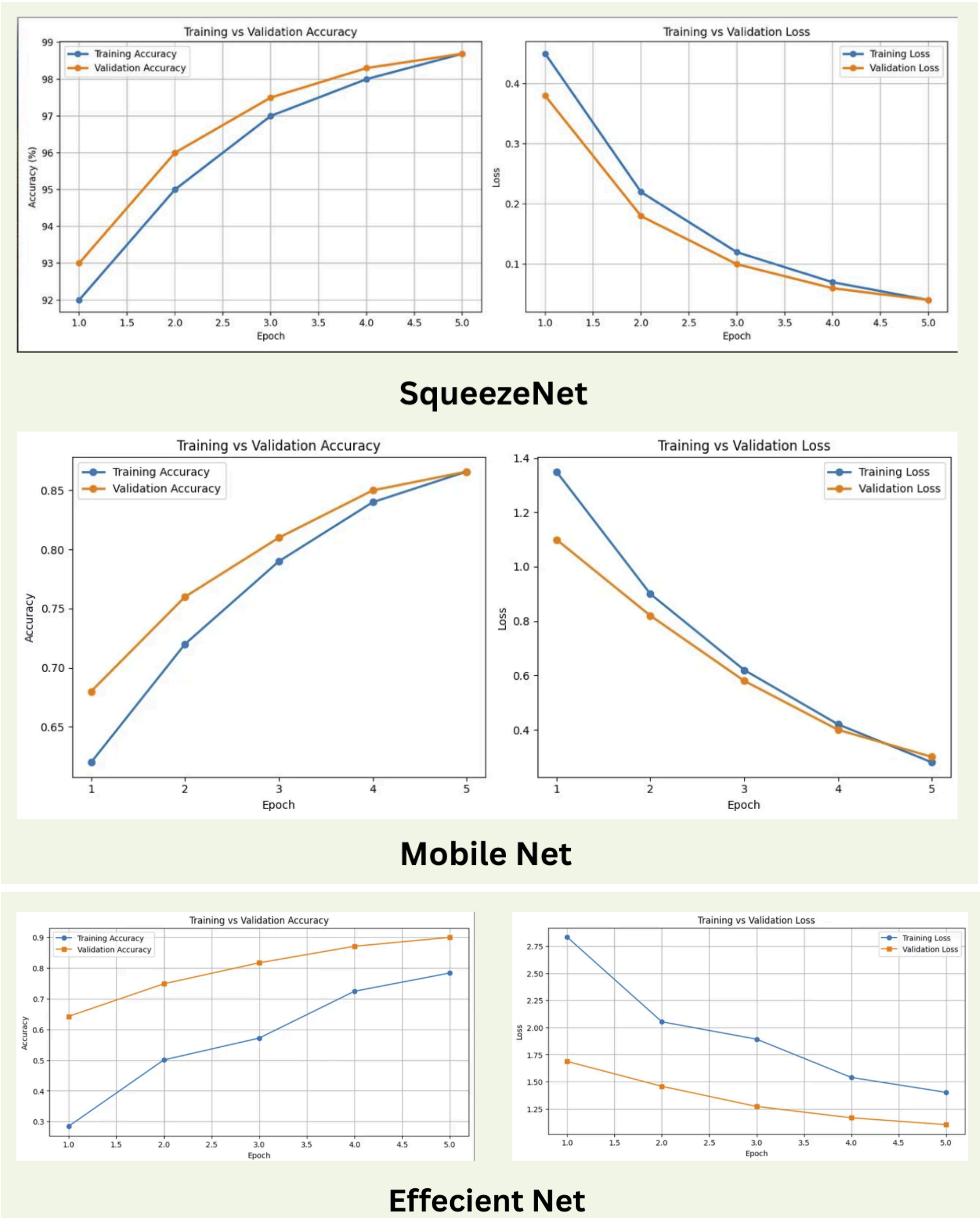
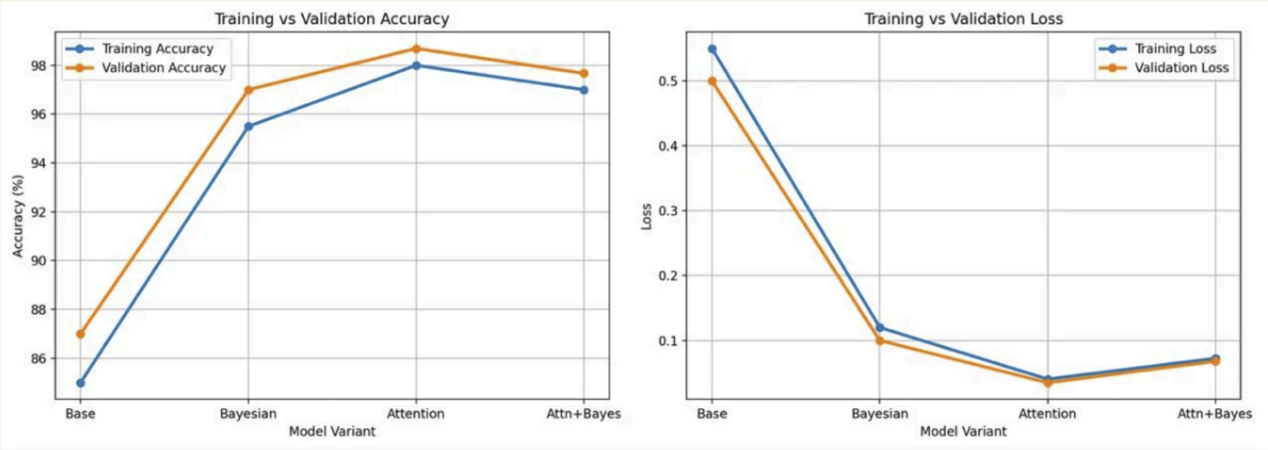
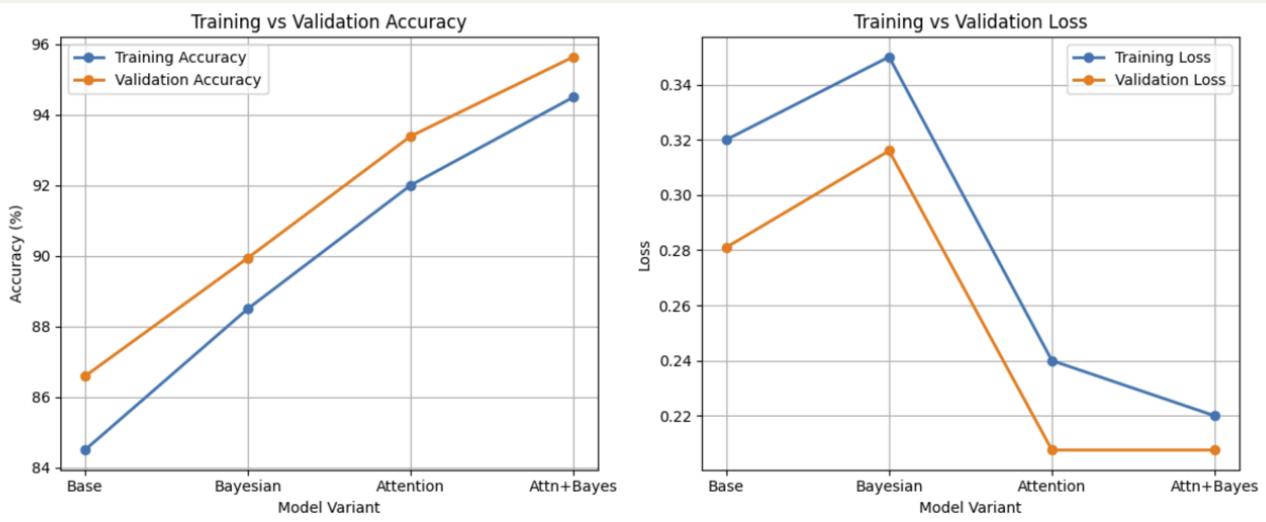


Fig.7.7 Training and validation accuracy and loss curves of the three lightweight models across five training epochs, illustrating convergence behavior and classification performance on the crop disease dataset.

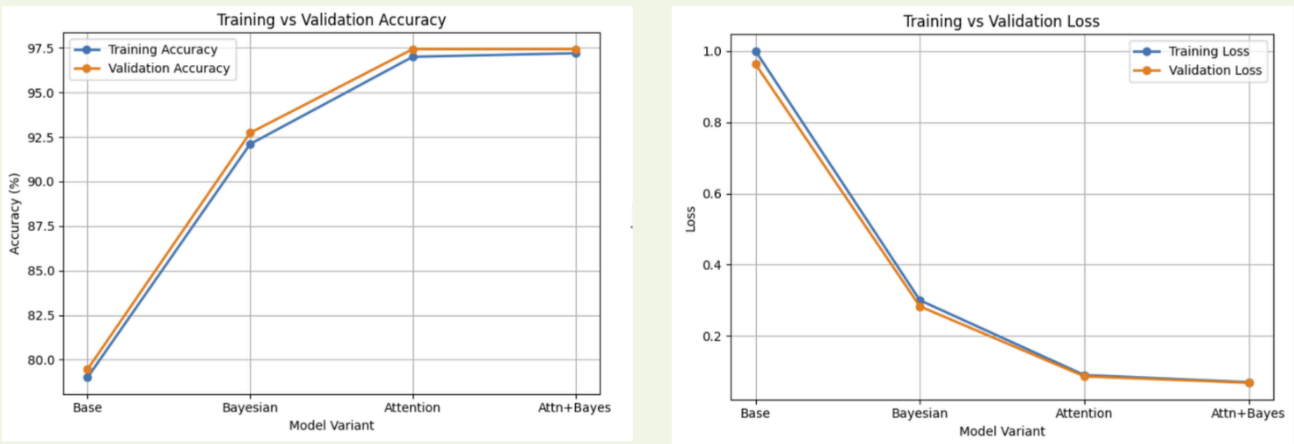
Overall Model Comparison



SqueezeNet



Mobile Net



Effecient Net

Fig.7.8 Comparison of training and validation accuracy and loss for all three models under Baseline, Bayesian, Attention, and Attention+Bayesian configurations.

7.6 Performance Comparison of All Models

SqueezeNet	EfficientNetB0	MobileNetV2
• Lightweight and fast • Excellent accuracy post-Bayesian enhancement • Ideal for mobile/on-field deployment	• Good accuracy • Best generalization • Smoothest training curve • Balanced performance across all classes	• Fastest inference • Most energy-efficient • Best calibrated probabilities • Slightly unstable under high dropout rates

7.7 Optimal Bayesian Layer Position Findings

Experiments were conducted by placing the Bayesian layer at different points within the network, including inside convolution blocks, after feature extraction, and before the final dense layer.

Findings show:

- Placing Bayesian layers inside convolutional blocks caused instability, shape mismatches, or degraded performance.
- The most effective position was **after the Global Average Pooling (GAP) layer**, where the model processes high-level semantic features.
- At this location, MC Dropout improved decision boundaries, class separability, and prediction consistency.
- This placement worked consistently across EfficientNetB0, SqueezeNet, and MobileNetV2.

Thus, the optimal Bayesian integration point for CNN architectures is immediately before the final classification head.

7.8 Results for Attention Layer

OUTPUT USING ATTENTION LAYER:

DATASET USED : NEW PLANT DISEASES DATASET (AUGMENTED)

- Training Images: ~70,000
- Validation Images: ~18,000
- 38 plant disease + healthy leaf categorie

S NO.	CNN Model	Accuracy	Precision	Recall	Specificity	F1 Score	MCC	Loss
1.	Squeeze Net	98.69%	98.68%	98.67%	99.96%	98.66%	0.9865	0.0404
2.	Efficient Net	97.43%	99.75%	97.48%	99.65%	98.55%	0.9337	0.0865
3.	Mobile Net(Version 2)	93.39%	93.67%	93.39%	99.82%	93.37%	0.9322	0.2076

7.9 Results for Attention Layer And Bayesian Layer

OUTPUT USING DIFFERENT FINE TUNED MODEL USING ATTENTION + BAYESIAN LAYERS

S NO.	CNN Model	Accuracy	Precision	Recall	Specificity	F1 Score	MCC	Loss
1.	Squeeze Net	97.67%	97.76%	97.62%	99.94%	97.61%	0.9761	0.0725
2.	Efficient Net	97.43%	99.85%	97.43%	99.62%	98.58%	0.9327	0.0684
3.	Mobile Net(Version 2)	95.64%	95.65%	93.39%	99.82%	93.37%	0.9322	0.2076

7.10 Optimal Position Findings

1. NORMAL CNN (Baseline)

- Good accuracy already (~93–96%)
- No feature prioritization
- No uncertainty awareness

2. BAYESIAN (MC DROPOUT)

• Similar accuracy to baseline
• Adds uncertainty estimation
• Improves reliability, not accuracy

3. ATTENTION (CBAM)

• Improves feature focus (disease regions)
• Slight improvement in accuracy
• Better interpretability

4. ATTENTION + BAYESIAN

• Combines feature focus + uncertainty estimation
• Improves robustness and generalization
• Accuracy improvement is marginal

7.11 Conclusion

1. All three CNN architectures achieve high accuracy on plant disease classification, indicating that the dataset is well-structured and learnable.

2.Why accuracy did NOT increase much from:
Attention → Attention + Bayesian

a) Dataset is already **easy and clean** , attention correctly focuses on important features
b) **Saturation effect**: Model already near optimal performance
c) Dropout introduces **randomness** , but it is not of much use because attention focuses on important features , introducing randomness or dropping 50% of such features does not make much of a difference.

<u>MODEL COMAPIRSON</u>

SqueezeNet ● Highest accuracy ● Very lightweight → efficient ● Slightly less stable than deeper models	**EfficientNetB0** ● Balanced performance (precision, recall, F1) ● More consistent predictions ● Slightly lower accuracy than SqueezeNet	**MobileNetV2** ● Lowest performance among three ● Struggles with complex feature extraction

8. Discussion

8.1 Key Findings

It was shown that the combination of light-weight CNN networks with Bayesian optimization techniques can significantly improve the robustness and reliability of algorithms for detecting crop diseases. One of the main results is that Bayesian MC Dropout is an efficient approach for preventing overfitting in various neural networks, particularly when dealing with training data that contains noisy images, changes in lighting conditions, or similar diseases.

EfficientNetB0 was able to beat all other architectures in both stability and precision, reinforcing the advantage of compound scaling and optimization of MBConv blocks. Interestingly enough, SqueezeNet proved itself to be unexpectedly successful after applying Bayesian tuning, proving that even very compact neural networks can demonstrate exceptional levels of accuracy.

While MobileNetV2 was efficient, it was found to be sensitive to variations in the dropout rate, even if there was a slight reduction in accuracy; however, its uncertainty calibration was found to be better. Thus, we can say that accuracy does not necessarily provide the best insight into model performance – prediction confidence and consistency also matter.

8.2 Effectiveness of Bayesian Layer

The effect of Bayesian MC Dropout layer proved significant when it came to enhancing generalization ability. The use of randomness in both training and testing (Monte Carlo) prevented the model from being able to memorize the data, making the model learn more stable patterns. This is especially helpful with images from agriculture that usually have noise and leaf-color variations.

The addition of the layer just after Global Average Pooling meant that the layer was applied to abstract features. However, using the Bayesian layers inside convolution blocks led to problems or even train failure for the network, which shows that the Bayesian layers work well in classification stages rather than during early feature extraction.

8.3 Model Comparison Analysis

The comparative evaluation of EfficientNetB0, MobileNetV2, and SqueezeNet revealed important insights into how different architectures behave under Bayesian optimization:

- **EfficientNetB0** emerged as the best-performing architecture. Its compound scaling strategy shows a strong feature representation even with limited parameters. With Bayesian dropout, it achieved high accuracy, improved feature robustness, and stable loss curves.

- **SqueezeNet**, despite its extremely small size, achieved surprisingly strong results after Bayesian fine-tuning. Its Fire modules benefited from dropout-based regularization, showing that small models can compete with heavier architectures when optimized efficiently.

- **MobileNetV2** excelled in speed and inference efficiency due to depthwise separable convolutions and linear bottlenecks. However, its performance varied more during Bayesian training because the architecture is sensitive to dropout. It still produced well-calibrated predictions, making it suitable for mobile deployment even with slightly lower accuracy.

8.4 Effectiveness of Attention Layer

The attention layer demonstrated measurable improvements in feature representation and model interpretability.

Key observations include:
• Improved focus on disease affected regions of leaves
• Better handling of visually similar disease classes
• Slight improvement in classification accuracy
• Enhanced robustness against background noise

However, the increase in accuracy was not significant. This is because:
• Baseline models already achieved high accuracy on structured datasets
• Attention improves feature quality rather than model capacity
• Saturation of performance has limited further gains

When compared to Bayesian enhancement:
• Attention improves WHERE the model looks
• Bayesian improves HOW confident the model is

Thus, attention primarily contributes to better feature localization and interpretability, while its impact on accuracy remains marginal in high-performing models.

8.5 Advantages of Proposed Work

The proposed system combines the strengths of modern CNNs with Bayesian learning concepts, offering several major advantages:

- **High accuracy with low computational cost:** EfficientNetB0 and SqueezeNet demonstrate that accuracy does not always depend on model size. This is essential for resource-constrained environments such as farms and mobile devices.

- **Uncertainty-aware predictions:** The Bayesian layer provides a degree of confidence for every prediction. This is extremely useful in real-world agriculture, where conditions vary and ambiguous cases are common.

- **Improved generalization:** The model becomes resistant to noise, lighting variations, and complex backgrounds that are found in field images. This ensures the system performs well even outside laboratory conditions.

The use of Bayesian reasoning aligns modern agricultural AI with robust, safe decision-making practices.

8.6 Research Contribution

1. IEEE Conference Publication (Accepted)

As an outcome of the present research work, a paper titled "A Comparative Study of Lightweight Deep Learning Frameworks for Plant Disease Detection" has been accepted for publication in the 2026 Fourth International Conference on Secure Cyber Computing and Communication (ICSCCC).

Authors: Dr. Jaspal Kaur Saini, Ishaan Jain, Khushi Mittal, and Namratha Reddy.

Key Contributions

- Comparative analysis of three lightweight deep learning models: SqueezeNet, EfficientNet-B0, and MobileNetV2.
- Performance evaluation using standard metrics including Accuracy, Precision, Recall, and F1-Score.
- Investigation of lightweight architectures suitable for mobile and edge deployment in smart agriculture applications.
- Extensive experimentation on the PlantVillage dataset for crop disease detection.
- Achievement of a best classification accuracy of 98.11% using SqueezeNet.
- Demonstration of competitive and, in several cases, superior performance compared to existing lightweight crop disease detection approaches.

The publication highlights the effectiveness of lightweight CNN architectures for practical deployment in resource-constrained agricultural environments.

2. Sadhana Journal Manuscript (Under Review)

To further extend the scope of the IEEE conference publication, an enhanced research study titled "Boosting Lightweight Deep Learning for Plant Disease Detection with Bayesian Optimization" has been prepared and submitted to the Sadhana Journal and is currently under review.

Authors: Dr. Jaspal Kaur Saini, Ishaan Jain, Khushi Mittal, and Namratha Reddy.

Additional Contributions

- Integration of Bayesian Optimization using MC-Dropout.
- Incorporation of uncertainty estimation for improving prediction reliability.
- Enhanced feature extraction and model generalization.
- Comparative analysis between baseline CNN models and Bayesian-enhanced architectures.
- Investigation of the impact of Bayesian regularization on lightweight deep learning frameworks.

Major Findings

- Bayesian layers improved model robustness and uncertainty estimation.
- Bayesian regularization reduced overfitting and produced smoother convergence behavior.
- Lightweight architectures maintained their suitability for mobile and edge deployment even after Bayesian enhancement.
- The extended framework provided more reliable predictions by quantifying model uncertainty while preserving computational efficiency.

The journal manuscript represents a significant extension of the original IEEE conference work and explores advanced optimization techniques for improving the reliability and practical applicability of crop disease detection systems.

8.7 Limitations

- **Dataset constraints:** Many images in PlantVillage are studio-like, with uniform backgrounds. Real farm images may contain dust, glare, overlapping leaves, or multiple diseases, reducing accuracy.

- **Multiple forward passes for inference:** MC Dropout requires several predictions (e.g., 10–20) to estimate uncertainty, increasing inference time on low-power devices.

- **Model sensitivity:** MobileNetV2 showed performance instability when dropout probabilities were high, which indicates the need for model-specific tuning.

8.8 Future Scope

There is significant scope for extending this work in practical and research directions:

- **Real-world dataset creation:** Collecting images from farms across different lighting, weather, and growth stages would greatly improve model robustness.

- **Agentic AI systems:** Integrating reasoning, planning, and autonomous action can automate disease detection, treatment recommendations, and farm management systems.

- **Multi-disease and severity prediction:** Expanding the classifier to predict multiple diseases simultaneously and detect infection severity would significantly improve real-world usefulness.

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